

Who Moved the Mode?

Gauge-Invariant Project-Cycle Responsibility for Dynamic-Security Externalities in Inverter-Dominated Power Networks

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Abstract—Three related questions arise when inverter-rich grids are planned with reduced models and multiple corrective projects: can the multivariable dynamics be reduced without deleting essential feedback structure, can individually screened actions be composed safely, and, when a harmful oscillatory mode is created jointly, which projects are responsible for moving it? Existing work addresses pieces of this chain—MIMO impedance coupling, topology and controller sensitivity, stability-preserving reduction, cooperative allocation, and graph-cycle identities—but does not provide a project-level spectral responsibility calculus that is invariant to arbitrary low-rank realizations of each project.

This synthesis develops that calculus and identifies the strongest publication path. Each project is represented as a finite-rank perturbation of a controller-aware nonlinear dynamic operator. The resulting project Green matrix is a block matrix whose off-diagonal blocks transmit one project's dynamic effect through the base grid to another. A change of basis inside any project acts by block similarity; perturbation determinants, coalition values, and traces of closed project walks are therefore representation invariant. The central step is to replace fixed-frequency log-determinant proxies by contour spectral moments. For a coalition S , the functional $\mathcal{M}_k(S; \Gamma) = \frac{1}{2\pi j} \oint_{\Gamma} s^k d \log \Delta_S(s)$ with $\Delta_S = \det T_S / \det T_0$ exactly equals the change in the k th spectral moment inside Γ . In particular, $k = 0$ counts created minus removed modes and, when one base and one perturbed mode are isolated, $k = 1$ equals the exact pole displacement. A finite Mobius transform then decomposes that displacement into intrinsic coalition dividends without any proxy correlation. In the rank-one certified region, each dividend is exactly the sum of prime intervention-cycle contributions having that coalition as support; pair responsibility collapses to the single feedback factor $1 - q_{ab}q_{ba}$. For block projects, the corresponding closed-walk traces are gauge invariant and provide the multivariable generalization.

The framework clarifies the role of Shapley allocation: Shapley is not the scientific novelty and has prior power-system use; it is an optional corollary that fairly splits already-derived physical coalition dividends. The strongest research program is therefore: (i) gauge-invariant project-cycle spectral responsibility (lead theory paper), (ii) coalition-aware proportional-impact rules for interconnection cluster studies, (iii) decision-preserving scalarization and minimal EMT-region certification, and (iv) minimum cycle surgery for mitigation. Existing IEEE 39-bus mechanism results already show the required phenomena—scalar connectivity can improve while a coupled mode destabilizes, two individually beneficial actions can jointly reduce damping by

10.76%, and interaction scores predict strong non-additivity—but the proposed exact mode-displacement attribution and project-block invariance require a dedicated full-order validation campaign.

Index Terms—inverter-based resources, dynamic security, small-signal stability, finite-rank perturbations, Green functions, spectral moments, cooperative games, Harsanyi dividends, Shapley value, graph zeta functions, holonomy, model reduction, interconnection clusters.

I. THE SCIENTIFIC GAP AFTER COMBINING THE THREE FRAMEWORKS

The companion manuscripts establish three distinct layers. The first asks whether a heterogeneous matrix-valued inverter network can be represented by independent scalar channels. Its answer is structural: after tree normalization, simultaneous diagonalizability of fundamental-cycle holonomies controls exact scalarizability, and a non-Abelian cycle defect quantifies one obstruction. The second asks whether corrective actions can be evaluated independently. Its answer is again structural: a finite-rank portfolio produces a small action Green matrix; acyclic action graphs compose exactly, while strongly connected components (SCCs) contain all determinant-level non-composability. The third asks how a jointly produced dynamic externality might be divided among projects and connects prime cycles to Harsanyi dividends and Shapley values.

The strongest remaining gap is not another scalar strength index, another first-order eigenvalue sensitivity, or another use of Shapley allocation. It is the missing bridge

$$\begin{aligned} \text{physical MIMO dynamics} &\rightarrow \text{project feedback cycles} \\ &\rightarrow \text{exact critical-mode displacement} \\ &\rightarrow \text{coalition responsibility.} \end{aligned} \tag{1}$$

This bridge is stronger than fixed-frequency interaction scores because it can attribute the actual movement of a critical mode, rather than a surrogate that merely correlates with damping non-additivity.

The novelty boundary must be explicit. Schur complements, resolvents, determinant lemmas, simultaneous diagonalization, graph zeta functions, Harsanyi dividends, Shapley values, Perron–Frobenius sensitivity, and the argument principle are established mathematics. Shapley-based small-signal-stability allocation has also appeared in power systems, including PSS ancillary-service pricing and game-theoretic identification

J. L. Mayorga Taborda is with Universidad de los Andes, Bogota, Colombia. This document is a definitive research synthesis prepared in August 2026. It consolidates the strongest publishable claims emerging from three companion manuscripts on scalarizability, intervention cycles, and cooperative attribution. Numerical evidence cited as mechanism-level uses reduced SG/GFL/GFM models; full-order DAE/EMT replication remains an explicit validation gate.

of stability-critical parameters [11], [12]. The publishable contribution is therefore the *physical identification* of project-coalition spectral dividends with closed feedback structures generated by the controller-aware grid resolvent, together with a representation-invariant project formulation and an exact contour-based attribution of mode count and mode displacement.

II. CONTROLLER-AWARE BASE OPERATOR AND PROJECT BLOCKS

A. Physical reduction

After algebraic elimination and a partition between retained interface states and internal controller/device states, the exact Schur reduction gives a nonlinear retained-space operator

$$\mathbf{T}_0(s) = s^2 \mathbf{M} + s \Sigma(s) + \mathbf{L}, \quad (2)$$

where $\Sigma(s)$ is generally matrix valued, non-diagonal, and frequency dependent. The purpose of (2) is not to claim a new Schur complement; it preserves controller memory in the planning operator instead of replacing all device effects by a fixed nodal damping coefficient.

B. Project-level finite-rank representation

Let project i —a line reinforcement, BESS/GFM installation, damping or inertia change, or controller retuning—produce a finite-rank operator perturbation

$$\Delta \mathbf{T}_i(s) = \mathbf{B}_i(s) \mathbf{D}_i(s), \quad \mathbf{B}_i \in \mathbb{C}^{n \times r_i}, \mathbf{D}_i \in \mathbb{C}^{r_i \times n}. \quad (3)$$

For a coalition S of projects, stack the factors into $\mathbf{B}_S$ and $\mathbf{D}_S$ so that

$$\mathbf{T}_S(s) = \mathbf{T}_0(s) + \mathbf{B}_S(s) \mathbf{D}_S(s). \quad (4)$$

Whenever $\mathbf{T}_0$ is invertible, define $\mathbf{G}_0 = \mathbf{T}_0^{-1}$ and the block project Green matrix

$$\mathbf{K}_S(s) = \mathbf{D}_S(s) \mathbf{G}_0(s) \mathbf{B}_S(s), \quad (5)$$

whose (i, j) block is

$$\mathbf{K}_{ij}(s) = \mathbf{D}_i(s) \mathbf{G}_0(s) \mathbf{B}_j(s). \quad (6)$$

The matrix determinant lemma gives

$$\Delta_S(s) \equiv \frac{\det \mathbf{T}_S(s)}{\det \mathbf{T}_0(s)} = \det(\mathbf{I} + \mathbf{K}_S(s)). \quad (7)$$

Thus all full-grid propagation is contained in $\mathbf{G}_0$, while the coalition is represented in a space whose dimension is the sum of project ranks.

C. Representation gauge inside each project

The factorization (3) is not unique. For any nonsingular $\mathbf{S}_i \in GL(r_i, \mathbb{C})$,

$$\mathbf{B}'_i = \mathbf{B}_i \mathbf{S}_i, \quad \mathbf{D}'_i = \mathbf{S}_i^{-1} \mathbf{D}_i \quad (8)$$

represents the same physical project. Let $\mathbf{S} = \text{blkdiag}(\mathbf{S}_i)$. Then

$$\mathbf{K}'_S = \mathbf{S}^{-1} \mathbf{K}_S \mathbf{S}. \quad (9)$$

Theorem 1 (Project-representation invariance). For every project coalition, the perturbation determinant (7), its zeros and poles, and every contour spectral moment defined in Section III are invariant under arbitrary internal project factorizations (8). Moreover, for any closed project walk $\gamma = (i_1 \rightarrow i_2 \rightarrow \dots \rightarrow i_\ell \rightarrow i_1)$,

$$\mathbf{H}_\gamma = \mathbf{K}_{i_1 i_2} \mathbf{K}_{i_2 i_3} \dots \mathbf{K}_{i_\ell i_1} \quad (10)$$

transforms by similarity at its base project; hence $\text{tr } \mathbf{H}_\gamma$, $\det \mathbf{H}_\gamma$, and $\text{spec}(\mathbf{H}_\gamma)$ are invariant.

Proof. Equation (9) follows by block substitution. The determinant in (7) is similarity invariant. For a closed walk, all intermediate factors $\mathbf{S}_i \mathbf{S}_i^{-1}$ cancel, leaving $\mathbf{H}'_\gamma = \mathbf{S}_{i_1}^{-1} \mathbf{H}_\gamma \mathbf{S}_{i_1}$. The stated invariants follow. $\square$

This theorem addresses a weakness of rank-one action graphs: responsibility must not depend on an arbitrary choice of internal columns used to factor a multi-state BESS or controller model. The project block, not an arbitrary rank-one column, is the physically meaningful player.

III. EXACT SPECTRAL RESPONSIBILITY: REPLACE THE PROXY BY THE POLE MOTION

A. Contour spectral moments

Let Γ be a positively oriented closed contour on which $\mathbf{T}_0$, $\mathbf{T}_S$, and Δ_S are nonsingular. Define

$$\mathcal{M}_k(S; \Gamma) = \frac{1}{2\pi j} \oint_\Gamma s^k \frac{\Delta'_S(s)}{\Delta_S(s)} ds, \quad k = 0, 1, 2, \dots \quad (11)$$

which avoids any global choice of logarithm branch.

Theorem 2 (Exact spectral-moment shift). Assume Δ_S is meromorphic inside Γ and nonzero on Γ . Then

$$\mathcal{M}_k(S; \Gamma) = \sum_{\lambda \in Z_S \cap \Omega} m_\lambda \lambda^k - \sum_{\nu \in P_S \cap \Omega} m_\nu \nu^k, \quad (12)$$

where Z_S and P_S are the zeros and poles of Δ_S , respectively, m denotes multiplicity, and Ω is the interior of Γ .

Proof. Apply the weighted argument principle to the meromorphic function $\Delta_S(s)$. $\square$

Because $\Delta_S = \det \mathbf{T}_S / \det \mathbf{T}_0$, the zeros correspond to coalition modes and the poles to base-system modes, modulo exact cancellations. Consequently,

$$\mathcal{M}_0(S; \Gamma) = N_S(\Omega) - N_0(\Omega), \quad (13)$$

$$\mathcal{M}_1(S; \Gamma) = \sum_{\lambda \in \mathbf{T}_S \cap \Omega} \lambda - \sum_{\lambda \in \mathbf{T}_0 \cap \Omega} \lambda. \quad (14)$$

If Γ encloses exactly one base mode λ_0 and its coalition continuation λ_S , then

$$\mathcal{M}_1(S; \Gamma) = \lambda_S - \lambda_0. \quad (15)$$

Thus $\text{Re } \mathcal{M}_1$ is the exact shift of decay rate and $\text{Im } \mathcal{M}_1$ is the exact shift of modal frequency. No correlation with a separate fixed-frequency determinant observable is required.

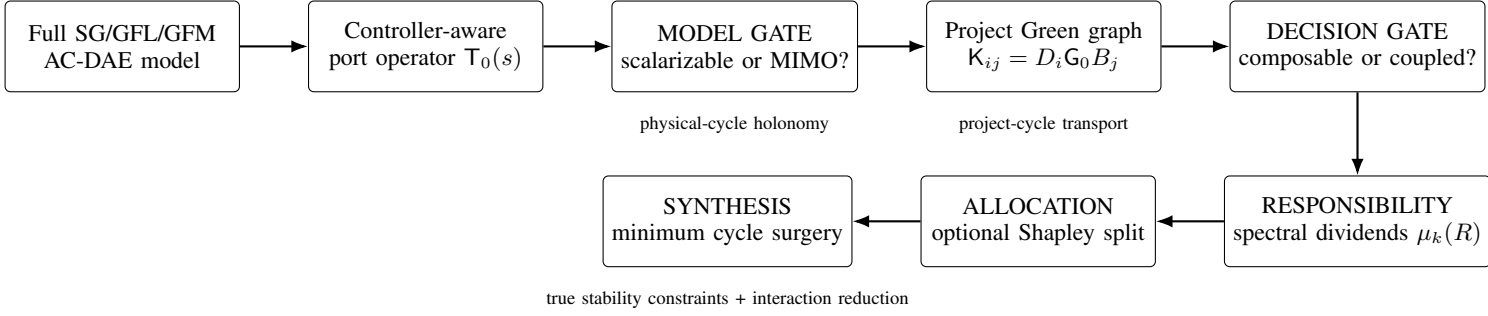


Fig. 1. Unified research architecture. The model gate asks whether scalar reduction is legitimate. The decision gate asks whether project actions compose. The new central layer attributes the resulting spectral motion to project coalitions; economics and mitigation are downstream uses, not substitutes for the physics.

B. Coalition spectral dividends

For a finite project set A , define the Mobius/Harsanyi transform of the spectral moment

$$\mu_k(R; \Gamma) = \sum_{S \subseteq R} (-1)^{|R|-|S|} \mathcal{M}_k(S; \Gamma). \quad (16)$$

By finite Mobius inversion,

$$\mathcal{M}_k(A; \Gamma) = \sum_{\emptyset \neq R \subseteq A} \mu_k(R; \Gamma). \quad (17)$$

Theorem 3 (Exact coalition responsibility for mode motion). For any contour satisfying Theorem 2, the complex spectral displacement of the coalition decomposes exactly into the intrinsic contributions (16). If (15) holds, then

$$\lambda_A - \lambda_0 = \sum_{\emptyset \neq R \subseteq A} \mu_1(R; \Gamma). \quad (18)$$

The real parts $\text{Re } \mu_1(R; \Gamma)$ sum exactly to the movement of the critical pole toward or away from instability, and the imaginary parts sum to its frequency shift.

The theorem is algebraically elementary once (11) is defined; the scientific contribution is the interpretation. A harmful oscillation need not have one source. A coalition can own a nonzero spectral dividend even when all of its proper subcoalitions are individually stabilizing.

C. Shapley is a corollary, not the novelty

If a regulator or planner requires project-level shares rather than coalition-level responsibility, apply Shapley separately to a real functional such as $v_\sigma(S) = \text{Re } \mathcal{M}_1(S; \Gamma)$:

$$\phi_i^{(\sigma)} = \sum_{R \ni i} \frac{\text{Re } \mu_1(R; \Gamma)}{|R|}. \quad (19)$$

This share is exact for the chosen spectral functional, but the equal split inside each dividend comes from the Shapley axiom package; it is not a physical law. This distinction removes an unnecessary novelty risk because Shapley-based stability allocation is established [11], [12].

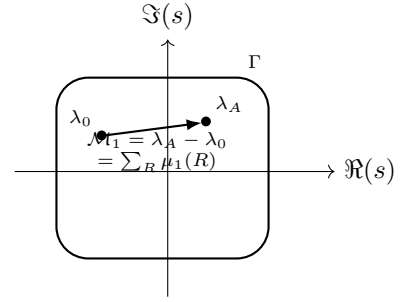


Fig. 2. Contour spectral responsibility. When one base mode and one coalition mode are isolated inside Γ , the first contour moment is exactly the pole displacement. Mobius inversion assigns that displacement to project coalitions without a fixed-frequency proxy.

IV. FROM SPECTRAL DIVIDENDS TO PROJECT FEEDBACK CYCLES

A. Rank-one projects: exact prime-cycle interpretation

For rank-one projects, the action Green matrix is scalar-valued. Split

$$K = K_d + K_o, \quad Q = (I + K_d)^{-1} K_o, \quad (20)$$

so that

$$\det(I + K) = \prod_i (1 + k_{ii}) \det(I + Q). \quad (21)$$

When $\rho(|Q(s)|) < 1$, the Bowen–Lanford/Ihara-type product for $\det(I + Q)$ is absolutely convergent:

$$\det(I + Q) = \prod_{\gamma \in \mathcal{P}} \left[1 - (-1)^{|\gamma|} q(\gamma) \right], \quad (22)$$

where $\mathcal{P}$ denotes prime closed cycles modulo rotation.

Theorem 4 (Prime-cycle spectral responsibility). Assume $\rho(|Q(s)|) \leq 1 - \delta$ uniformly on Γ . For any coalition R with $|R| \geq 2$,

$$\mu_k(R; \Gamma) = \sum_{\substack{\gamma \in \mathcal{P} \\ \text{supp}(\gamma) = R}} \frac{1}{2\pi j} \oint_{\Gamma} s^k d \log \left[1 - (-1)^{|\gamma|} q(\gamma, s) \right]. \quad (23)$$

The series is absolutely and uniformly convergent under the stated certification, so the support grouping and contour integral may be interchanged.

For a pair $R = \{a, b\}$, the only prime cycle on exactly two scalar projects is the alternating two-cycle, giving

$$\mu_k(\{a, b\}; \Gamma) = \frac{1}{2\pi j} \oint_{\Gamma} s^k d \log(1 - q_{ab}q_{ba}). \quad (24)$$

Equation (24) is a sharper statement than a pairwise interaction score: it assigns the actual contour spectral moment of the pair synergy to the closed feedback product $q_{ab}q_{ba}$.

B. Block projects: gauge-invariant closed-walk bundles

For multivariable projects, a scalar Euler factor per edge is not representation invariant. The robust generalization is the trace expansion. Let Q be block-normalized, with diagonal blocks removed. In a region where $\rho(Q) < 1$,

$$\log \det(I + Q) = \sum_{p=2}^{\infty} \frac{(-1)^{p+1}}{p} \text{tr}(Q^p). \quad (25)$$

Expanding the block trace gives

$$\text{tr}(Q^p) = \sum_{i_1, \dots, i_p} \text{tr}(Q_{i_1 i_2} Q_{i_2 i_3} \cdots Q_{i_p i_1}). \quad (26)$$

Each summand is the trace of a project-cycle holonomy and is invariant by Theorem 1. Grouping terms by the exact set of project indices visited yields a representation-invariant support decomposition. Recent quiver determinant identities provide a broader mathematical setting for such block-cycle trace formulas [6]; the power-system contribution is the project-responsibility interpretation.

Proposition 1 (Block support responsibility). Under uniform convergence of (25) on Γ , the interaction part of each coalition spectral dividend can be written as a sum of contour moments of gauge-invariant closed project-walk traces whose exact support is that coalition. The finite Mobius definition (16) remains valid even when the cycle series is not certified; only the cycle representation is then withheld.

This separation is important. Exact spectral responsibility does not require $\rho(|Q|) < 1$; the convergence condition is only a gate for interpreting the result through a certified cycle expansion.

V. HOW THE SCALARIZABILITY THEORY STRENGTHENS THE RESPONSIBILITY THEORY

The physical network and the project graph are different objects. The scalarizability manuscript works on the physical port graph: local coordinate changes act on device port spaces and fundamental-cycle holonomies decide whether one common scalar basis exists. The responsibility theory works on the project graph: internal project realizations act by block similarity and closed project-walk holonomies identify representation-invariant feedback among interventions.

The two theories therefore form consecutive gates rather than duplicate one another. **Model gate:** can the physical MIMO dynamics be scalarized? **Decision gate:** can the project interventions be composed independently? **Responsibility layer:** which coalitions moved the critical spectrum?

A reduced model $\hat{T}_0$ induces $\hat{G}_0 = \hat{T}_0^{-1}$ and project interaction matrix $\hat{K} = D\hat{G}_0B$. The resolvent identity gives

$$G_0 - \hat{G}_0 = -G_0(T_0 - \hat{T}_0)\hat{G}_0, \quad (27)$$

so

$$\|K - \hat{K}\| \leq \|D\| \|G_0\| \|T_0 - \hat{T}_0\| \|\hat{G}_0\| \|B\|. \quad (28)$$

This elementary bound exposes a publishable third question: a model can reproduce the base-grid stability verdict yet fail to preserve the ranking or interaction structure of candidate interventions. A future decision-preserving reduction certificate should propagate a physical scalarization/model error through (28) to a bound on coalition ranking or contour spectral responsibility.

VI. THE FIXED-BASE LIMITATION AND THE EQUILIBRIUM-CONSISTENT EXTENSION

The clean principal-submatrix structure of K_S assumes a common base operator and additive finite-rank project perturbations. Real grid upgrades can also change the AC equilibrium. If the operating point satisfies

$$F(x^*, \theta) = 0, \quad (29)$$

then, whenever F_x is nonsingular,

$$\frac{\partial x^*}{\partial \theta_i} = -F_x^{-1} F_{\theta_i}. \quad (30)$$

For a dynamic operator $T(s, x^*, \theta)$,

$$\frac{dT}{d\theta_i} = \underbrace{T_{\theta_i}}_{\text{direct/structural}} - \underbrace{T_x F_x^{-1} F_{\theta_i}}_{\text{equilibrium mediated}}. \quad (31)$$

The existing controller-aware topology derivative further separates structural, power-flow-mediated, and controller-mediated channels.

This matters for attribution. With full AC re-solution, T_S is not generally obtained by simply deleting rows and columns from one fixed K_A . Therefore the claim that one G_0 exactly prices every realistic coalition must remain restricted to a fixed-base study snapshot. The stronger, publishable extension is an *equilibrium-consistent coalition game*: evaluate $\mathcal{M}_k(S; \Gamma)$ after re-solving each coalition equilibrium, then use the same finite Mobius transform (16). The decomposition remains exact as a set-function identity, but the fast principal-submatrix shortcut no longer applies.

A useful research target is to separate

$$\mu_k(R) = \mu_k^{\text{feedback}}(R) + \mu_k^{\text{equilibrium}}(R) + \mu_k^{\text{higher}}(R), \quad (32)$$

where the first term is predicted by the fixed-base project-cycle calculus and the second is induced by equilibrium migration. Equation (32) is a candidate theorem/program, not yet an established result of the existing mechanism experiments.

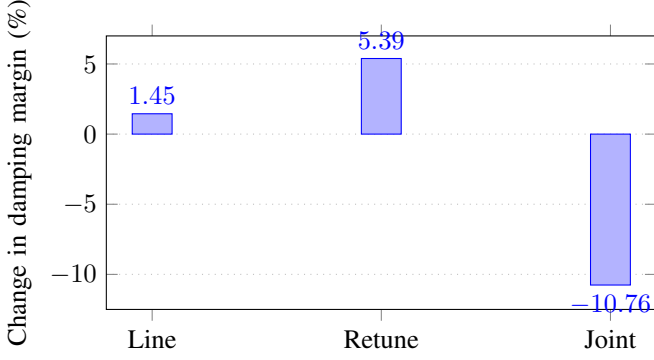


Fig. 3. Flagship mechanism-level non-composability result already established in the intervention-cycle manuscript. The new target is to decompose the exact critical-pole displacement behind this bar chart, rather than use a fixed-frequency proxy.

VII. MINIMUM CYCLE SURGERY: FROM ATTRIBUTION TO PRESCRIPTIVE DESIGN

Attribution is most valuable when it identifies the smallest modification that removes the harmful interaction. Let θ collect tunable controller and project parameters. A correct design problem must use a true dynamic-security constraint,

$$\operatorname{Re} \lambda_{\text{crit}}(\theta) \leq -\sigma^* \quad \text{or} \quad \zeta_{\min}(\theta) \geq \zeta^*, \quad (33)$$

plus operating constraints, not merely the cycle-series convergence condition.

For rank-one interaction cores, define the nonnegative majorant $|Q_c(\theta)|$ and $\bar{q} = \rho(|Q_c|)$. If the core is irreducible and u, v are its Perron left/right eigenvectors with $u^T v = 1$, then

$$\frac{\partial \bar{q}}{\partial |q_{ab}|} = u_a v_b. \quad (34)$$

The ratio $u_a v_b / c_{ab}$ ranks edges by first-order reduction in interaction radius per unit retrofit cost. The planning problem should therefore combine (33) with an interaction regularizer or constraint such as

$$\rho(|Q_c(\theta)|) \leq \eta < 1, \quad (35)$$

when a certified low-order cycle description is desired. Importantly, $\rho(|Q|) = 1$ is a convergence boundary for the cycle expansion, not a power-system stability boundary.

A stronger future version can target the exact harmful dividend directly:

$$\min_{\Delta\theta} C(\Delta\theta) \quad \text{s.t.} \quad \operatorname{Re} \mu_1(R^*; \Gamma, \theta + \Delta\theta) \leq \epsilon, \quad (36)$$

while maintaining (33). This changes “cycle surgery” from a heuristic graph operation into optimization against an exact spectral responsibility object.

VIII. EVIDENCE ALREADY AVAILABLE FROM THE COMPANION CAMPAIGNS

The new spectral-moment theory has not yet been simulated directly. However, the three companion campaigns establish the physical phenomena it is intended to explain. Table I separates existing evidence from new claims.

IX. A REAL INTERCONNECTION-STUDY USE CASE, WITH THE CLAIM NARROWED CORRECTLY

FERC Order No. 2023 requires cluster studies and a proportional-impact method for allocating system network upgrade costs within a cluster [14], [15]. This does not imply that current practice is devoid of technical impact methods. For thermal and voltage constraints, mature flow/impact calculations exist. Dynamic constraints are more difficult because project effects can be non-additive.

A concrete public example is MISO’s documented transient-stability allocation procedure: projects causing instability are identified by backing them out one at a time, and if multiple projects are identified, the transient-stability-driven network-upgrade cost is allocated pro rata by their total MW requests [16]. This procedure is not the same as the small-signal converter problem considered here, and the present framework should not claim to solve large-disturbance transient stability without further work. It does, however, expose a closely related technical gap: if two inverter projects are individually benign but jointly create an oscillatory mode, one-at-a-time causation identifies the participants but does not measure the interaction term that made the coalition harmful.

The proposed proportional-impact object for oscillatory small-signal security is therefore not MW and not Shapley itself. It is the coalition spectral dividend $\operatorname{Re} \mu_1(R; \Gamma)$ (or μ_0 for a created-mode count). An economic rule can then map the technical responsibility to dollars. That policy layer must remain separate from the physical theorem.

X. THE FOUR STRONGEST PUBLICATION OPTIONS

Combining the three manuscripts suggests four papers that are more defensible than a single oversized article. Table II ranks them by originality, maturity, industrial relevance, and review risk.

A. Paper 1: the lead theoretical paper

The recommended title is

Who Moved the Mode? Gauge-Invariant Project-Cycle Attribution of Dynamic Spectral Externalities in Multi-Project Power Networks.

Its publishable center should be Theorems 1–4, not Shapley. The critical numerical experiment is a 10–20 project cluster in which the exact real-part pole displacement is reconstructed from coalition dividends to machine precision, then further interpreted through certified cycle bundles. The decisive invariance test is to randomly refactor every multi-rank project and verify identical attribution.

B. Paper 2: the cluster-study paper

This paper should use realistic multi-owner line/BESS/GFL/GFM clusters and compare:

- 1) one-at-a-time back-out causation;
- 2) MW pro rata;
- 3) marginal eigenvalue sensitivity;
- 4) brute-force Shapley on the true full-model margin;
- 5) the proposed spectral-dividend responsibility;

TABLE I
EXISTING MECHANISM-LEVEL EVIDENCE AND WHAT IT SUPPORTS

Study element	Reported result	What is established	What remains to validate
Scalarizability / cospectral test	2,000 Laplacian-cospectral topology pairs; scalar spectra agree to 7.3×10^{-14} ; 4.35% have opposite full-MIMO stability verdicts	Scalar topology and unchanged local coupling statistics are not sufficient for the MIMO stability verdict	Reproduce with full manufacturer-grade GFL/GFM models and larger networks
IEEE 39-bus connectivity reversal	Line 25–38: GFV +0.1459%, Fiedler +0.2980%; critical mode moves from $-0.0151 + j17.2259$ to $+0.2502 + j19.7789$; decoupled model misses the event	Improved scalar connectivity does not universally order the coupled spectrum; reduced model can miss a controller-network mode	EMT/DAE full-order replication and independent implementation
Targeted mitigation	Retuning PLL proportional gain at bus 30 to $K_p = 3$ restores the new-line mode to $-0.1006 + j19.6263$	A remote controller can repair a topology-induced coupled mode without removing the line	Show that spectral responsibility identifies the same controller and that responsibility-guided tuning outperforms brute force
Mixed line/controller portfolio	Line 26–29 alone: +1.45% damping margin; controller retune alone: +5.39%; joint action: -10.76%	Individually beneficial actions can be jointly harmful	Attribute the exact pole displacement of this case by $\mu_1(R; \Gamma)$
Large pair campaigns	17,664 line/controller pairs: fixed-frequency spectral interaction vs damping non-additivity Spearman 0.673, AUC 0.897 for top 5%; 23,000 GFM-like pairs: Spearman 0.747, AUC 0.894	Action Green geometry predicts where non-additivity is strong	Replace proxy correlation by exact contour responsibility and test ranking fidelity
Topology-only portfolios	Pair interaction kernel Spearman 0.772 with Mobius damping interaction, AUC about 0.96; triple kernel Spearman 0.874 with genuine third-order interaction	Higher-order action cycles carry useful information beyond isolated ranking	Verify exact spectral dividends for pairs/triples and compare with direct damping-game dividends
Low inertia/damping scaling	Co-retirement slopes approximately -0.989 (pair), -1.469 (triple); fixed absolute damping reverses signs	Interaction strength can grow with joint loss of inertia and damping under isolated-mode assumptions	Extend to multimode, non-normal, controller-frequency-dependent cases

6) responsibility-guided mitigation cost.

The central engineering example should include a coalition that is harmful only through interaction, so that a static MW share is demonstrably disconnected from the spectral mechanism.

C. Paper 3: decision-preserving model reduction

The key falsification experiment is not “the reduced model has eigenvalue error.” It is

base-grid verdict correct, intervention decision wrong. (37)

For example, the full and reduced models can both classify the base grid as stable, while the reduced model ranks portfolios $A > B > C$ and the full MIMO model ranks $C > A > B$ or finds $A + B$ unsafe. A valid certificate should identify that the base study is trustworthy but the portfolio decision is outside the decision-preserving region.

D. Paper 4: surgery

The design paper begins where the responsibility paper ends. It should report the minimum controller/network change that reduces the identified harmful dividend while preserving the beneficial network upgrade. The existing single-PLL repair on IEEE 39 buses is a useful precursor but not sufficient evidence by itself.

XI. FULL-ORDER VALIDATION CAMPAIGN NEEDED BEFORE STRONG CLAIMS

A Transactions-grade campaign should be organized around falsification rather than volume. The following six tests are sufficient to establish or appropriately narrow the theory.

A. E1: project-factorization invariance

Use multi-rank GFL, GFM, BESS, and line/project blocks. Generate hundreds of random invertible internal transformations S_i in (8). Verify identical perturbation determinants, spectral dividends, and closed-walk trace invariants to numerical precision.

B. E2: exact mode-displacement conservation

For each random project coalition, track one isolated critical mode with a contour Γ and verify

$$\left| \lambda_S - \lambda_0 - \sum_{R \subseteq S, R \neq \emptyset} \mu_1(R; \Gamma) \right| \quad (38)$$

near machine precision. This is the new flagship result and should replace proxy-correlation plots as the principal validation.

TABLE II
RECOMMENDED PUBLICATION PORTFOLIO AFTER THE COMBINED NOVELTY AUDIT

Paper	Central claim	Originality	Maturity	Industrial value	Risk
1. Who Moved the Mode? Gauge-Invariant Project-Cycle Spectral Responsibility	Exact contour spectral-moment displacement is decomposed into representation-invariant project-coalition dividends; rank-one dividends admit prime-cycle support interpretation	9.8/10	7.0/10	9.3/10	Medium
2. Coalition-Aware Proportional Impact for Dynamic-Security Cluster Studies	Replace one-at-a-time/MW allocation of non-additive oscillatory interactions by a physics-grounded coalition responsibility object; Shapley is optional downstream allocation	9.1/10	6.5/10	10/10	Medium
3. Decision-Preserving Scalarization and Minimal Coupled-Model Certification	A reduced model is certified not only to preserve stability, but to preserve intervention ranking and coalition decisions; identifies the minimum MIMO/EMT study region	9.3/10	6.0/10	9.5/10	Medium-high
4. Minimum Cycle Surgery for Converter-Dominated Planning	Attribute the harmful feedback core, then compute the least-cost controller/network change that removes its exact spectral dividend while satisfying true stability constraints	8.9/10	6.0/10	9.6/10	Medium

C. E3: prime-cycle correspondence

For certified rank-one or scalarized project sets, compute the same pair/triple spectral dividends using (i) direct coalition Mobius inversion and (ii) prime-cycle factors. Test the truncation certificate versus actual error as the interaction radius approaches one.

D. E4: fixed-base versus re-equilibrated attribution

For line and BESS projects, compare fixed-base cycle responsibility with full AC re-solved coalition attribution. Quantify the residual and test whether (32) separates direct feedback from equilibrium-mediated interaction.

E. E5: cluster proportional-impact benchmark

Build multi-owner project clusters with matched MW ratings but different feedback geometry. Compare cost/responsibility shares from MW pro rata, one-at-a-time removal, margin-game Shapley, and spectral dividends. The strongest case is one where MW-identical projects have sharply different spectral responsibility.

F. E6: responsibility-guided surgery

For every destructive SCC, compare exhaustive retuning, local eigenvalue-sensitivity retuning, Perron-targeted interaction reduction, and direct spectral-dividend minimization. Report achieved damping margin, parameter cost, number of full-model evaluations, and whether the beneficial network upgrade can be retained.

The minimum external validation set should include the IEEE 39-bus benchmark with documented GFL/GFM models, a larger 68- or 118-bus system, and at least one EMT or manufacturer-grade converter implementation. A failed full-order reproduction should narrow the engineering claims without invalidating the exact finite-dimensional determinant and spectral-moment identities.

XII. CLAIMS READY NOW, CLAIMS THAT NEED DOWNGRADING, AND CLAIMS WORTH PURSUING

XIII. RECOMMENDED FINAL SCIENTIFIC NARRATIVE

The cleanest long-term story is a four-stage theory:

Stage 1: Model validity. A heterogeneous inverter grid is a matrix-valued dynamical graph. Cycle holonomies determine whether scalar channels exist or whether MIMO structure must be retained.

Stage 2: Decision interaction. Candidate upgrades form a second graph in project space. Acyclic project graphs compose; SCCs are the feedback cores that require joint study.

Stage 3: Spectral responsibility. Contour spectral moments assign the actual movement or creation of critical modes to project coalitions. In certified scalar/rank-one regions, those coalition dividends are prime feedback-cycle bundles; in the multivariable case they are gauge-invariant closed-walk bundles.

Stage 4: Prescriptive planning. Responsibility is used either for technical proportional-impact allocation or for minimum-cost surgery that removes the harmful coalition while retaining useful upgrades.

This narrative avoids the two weakest framings in the current drafts: “we invented new matrix mathematics” and “Shapley tells us causation.” The scientific statement is narrower and stronger: *the physical dynamic operator induces a representation-invariant coalition decomposition of critical spectral motion; game-theoretic allocation and retrofit decisions are downstream consequences.*

XIV. CONCLUSION

The combined theory is most valuable when it stops treating scalarizability, intervention non-composability, and cost attribution as separate topics. They are successive questions about information loss and responsibility. The scalarizability

TABLE III
CLAIM CALIBRATION FOR THE DEFINITIVE RESEARCH PROGRAM

Claim	Status	Recommended wording
Schur/self-energy controller-aware reduction	Established tool / validated implementation	Use as the master modeling bridge; do not claim the mathematics as new
Physical cycle scalarizability criterion	Candidate original power-system formulation	Claim network-level structural certification of exact scalar-channel representability, not invention of simultaneous diagonalization or holonomy
Action/project portfolio determinant and SCC composability	Candidate original formulation	Claim a project-space dynamic planning formulation and exact composability consequences
Shapley allocation for stability	Not new	Cite prior PSS/stability game-theory literature; use only as a downstream fair-allocation corollary
Fixed-frequency Euler factor “equals damping damage”	Not supported	Do not claim; current evidence is correlational
Contour first moment equals isolated critical-pole displacement	Exact mathematics once contour hypotheses hold	Make this the new central responsibility functional; validate numerically on full models
Mobius decomposition of spectral displacement	Exact finite set-function identity	Claim exact coalition spectral responsibility; novelty is its use on project-induced dynamic modes and connection to feedback cycles
Prime-cycle support equals rank-one spectral dividend	Candidate central theorem under certified convergence	Strong lead-paper claim after formal proof and direct numerical verification
Project-block gauge invariance	Candidate central theorem	Strong lead-paper claim; essential to avoid dependence on arbitrary low-rank realization
One base Green matrix prices every realistic coalition	Too strong	Restrict to fixed-base additive perturbations; develop equilibrium-consistent extension separately
$\rho(Q) = 1$ is a stability boundary	False	State only as a cycle-expansion/composability certification boundary
Low-inertia cycle scaling is universal	Too strong	Restrict to the derived isolated-mode/co-retirement assumptions until multimode validation

layer decides whether a reduced model has discarded essential matrix feedback. The project Green layer decides whether proposed actions can be screened independently. The new spectral-responsibility layer then answers the more difficult question: which project coalitions actually moved or created the critical mode?

The central recommendation is to replace the fixed-frequency log-determinant game as the headline contribution by the contour spectral-moment game. This eliminates the main proxy gap. For an isolated mode, the first contour moment is exactly its pole displacement; finite Mobius inversion then assigns that displacement to coalitions exactly. In rank-one certified regions, each coalition dividend is a bundle of prime intervention cycles with that exact support. In the block-project formulation, closed-walk holonomies are invariant to arbitrary internal project realization, preventing attribution from depending on modeling coordinates. Shapley allocation remains useful, but only after the physical dividends have been derived.

The strongest paper is therefore *Who Moved the Mode?* The strongest industrial follow-on is a coalition-aware proportional-impact method for interconnection cluster studies. Decision-preserving reduction and minimum cycle surgery form natural third and fourth papers. The current mechanism-level results are already sufficient to motivate all four. The decisive next step is not more rhetoric or more low-order algebra; it is a full-order campaign that verifies exact mode-displacement conservation, representation invariance, equilibrium effects, and responsibility-guided mitigation.

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